

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Vikas Gautam	Bangalore (India)	vikasgtm@gmail.com	
José Martín Muñoz	Madrid (Spain)	13josemart@gmail.com	
Saurav	India	sauraviiser@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

Team member 1	Vikas Gautam
Team member 2	Jose Martin
Team member 3	Saurav

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

Group WORK Project 1

Step 2

2.a. What is the problem the student's thesis attempts to solve?

The main problem that the thesis tries to tackle is how to forecast the price of crude oil in a more accurate and realistic way. Crude oil prices are influenced by tons of factors, from global politics and economic indicators to production decisions by OPEC. Traditional models, like GARCH for example, often ignore these complexities and just rely on past price movements. That kind of model doesn't really reflect how messy and interconnected the real world is.

The thesis proposes a more intelligent approach: using Probabilistic Graphical Models (PGMs), specifically Bayesian Networks and Hidden Markov Models, to model the oil market. It doesn't just look at historical prices, but also tries to understand how different variables influence each other and how uncertainty plays a role.

Three main issues are highlighted:

1. Traditional models like GARCH don't account for macro or geopolitical variables (like a war or a major policy change), which clearly affect oil prices.
2. Manual analysis is time-consuming and error-prone, especially when trying to connect different datasets from multiple sources.
3. There's also the need to connect futures prices with spot price forecasts, since futures reflect market expectations, and that's very useful information to include.

In short, the thesis is about creating a smarter, data-driven model that learns from both data and domain knowledge, in order to improve predictions and better understand what drives oil prices.

2.b. Why are Bayesian networks well suited to solve it?

Bayesian networks are a really good fit for this kind of problem because they don't just look at data blindly, they try to understand the relationships between variables — and that's exactly what you need when dealing with something as complex as the oil market.

First off, they're causal models, which means they can show how one thing influences another. For example, higher interest rates might lower economic growth, which then affects oil demand. These types of connections are really important and would just be ignored in most time-series models.

Another big reason is that BNs can handle missing or uncertain data. In the real world, we often don't have all the data we want, or some of it is delayed. Bayesian networks don't freak out in those cases — they can still make predictions using probabilities based on what's known.

Also, they're flexible. If you want to add a new variable (like oil tanker traffic data, or some satellite-based measurement), you don't need to start the model from scratch. You just plug it into the network and define how it connects with the other nodes.

Another important point is that BNs let you combine expert knowledge with actual data. So if someone who knows the industry tells you that OPEC decisions usually affect prices within 1-2 months, you can build that into the model, and then the machine can figure out the numbers based on the data.

Finally, Bayesian networks are transparent. You can explain how a prediction was made, which is super valuable in risk management or policymaking. People don't just want the number, they want to know why the model made that call.

2.c. What are the advantages of using this methodology for this problem?

There are quite a few. First, oil prices are notoriously tricky because they depend on so many things: inflation, wars, production quotas, economic growth, even rumors sometimes. A Bayesian network helps make sense of this mess by organizing these variables into a structured model, showing how they affect each other.

Second, it's common to run into missing data or delayed indicators, especially for macroeconomic stats or political events. A BN doesn't require all variables to be present, it can still generate useful predictions from partial data, which is a huge plus.

Third, you can bring in expert input, like saying "we know that political instability in the Middle East usually impacts oil prices." You don't have to rely 100% on past data, which might not reflect future situations anyway.

Also, these models are modular and scalable. Say a new data source becomes available, or a new factor becomes relevant (like the COVID pandemic or a new conflict). You don't have to throw everything away and start over. Just update the network.

Another benefit is that it can help improve forecast accuracy, especially when markets are under stress or behaving abnormally. Traditional models often break during crises, while Bayesian networks can adjust better because they're based on relationships, not just past price behavior.

And maybe one of the most practical advantages: they're explainable. You can say exactly which factors led to a certain forecast. That's super useful if you're a trader, a policymaker, or someone managing risk.

Lastly, one cool thing the thesis points out is that futures prices, which reflect how traders expect the market to move, can be used directly in the model. Instead of guessing what people think, you can include it in the network as a predictive signal.

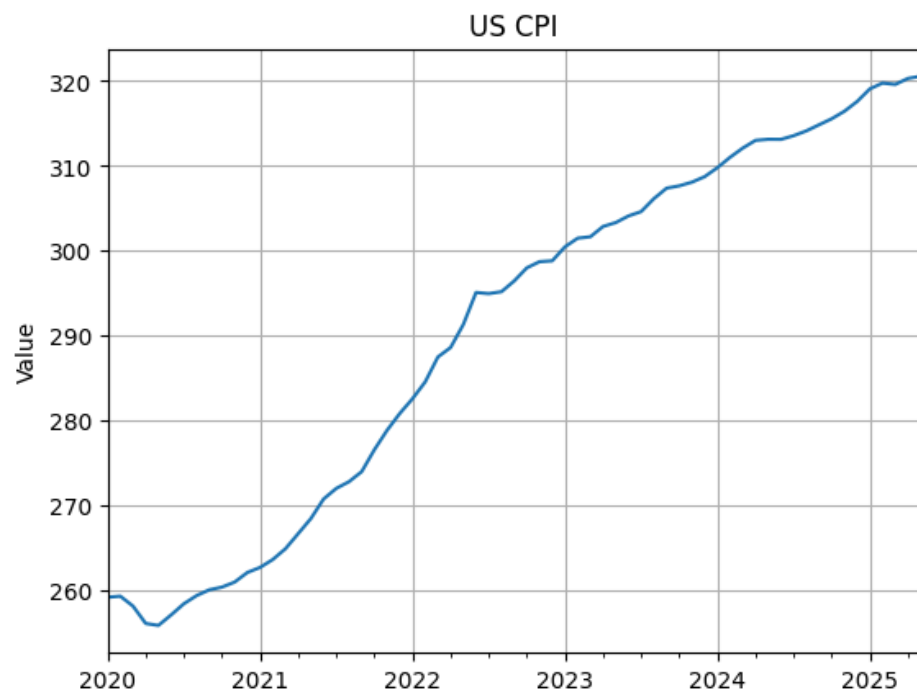
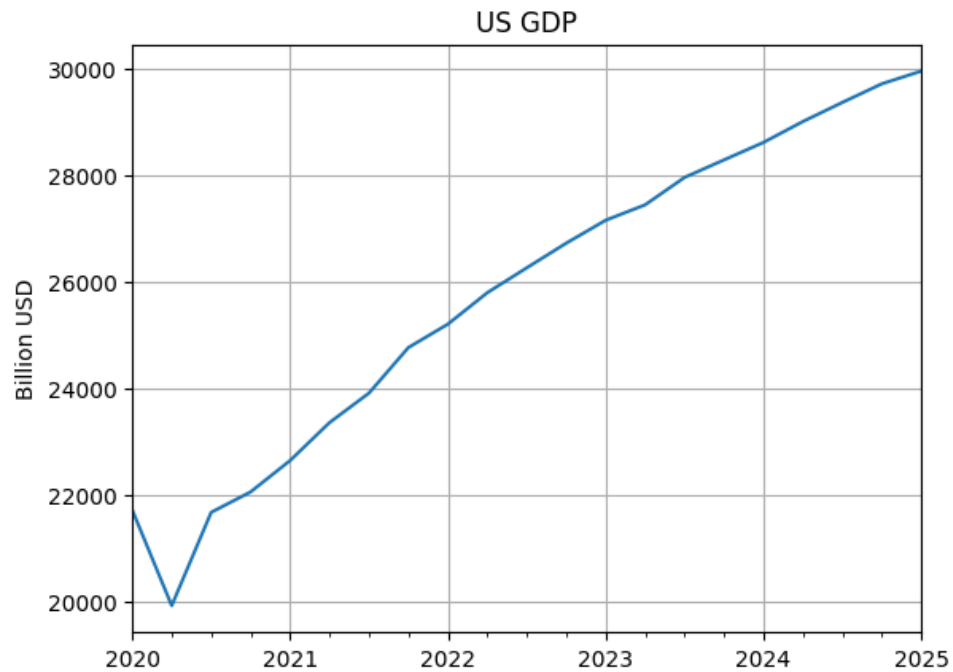
Step 3

We imported data from 2020-01-01 to 2025-05-30 from sources like yahoo finance or Fred using api .

Student A: Macroeconomic / Geopolitical Specialist

Dataset Imported:

- US Real GDP (GDP)
- Consumer Price Index (CPI) (CPIAUCSL')
- Source: FRED API

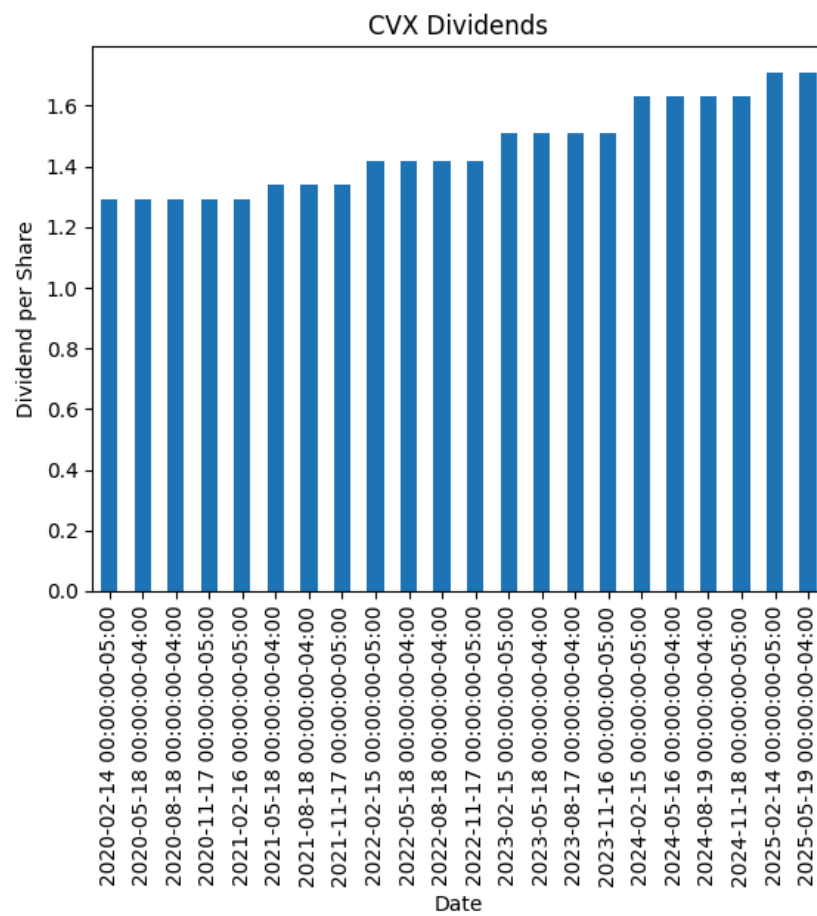
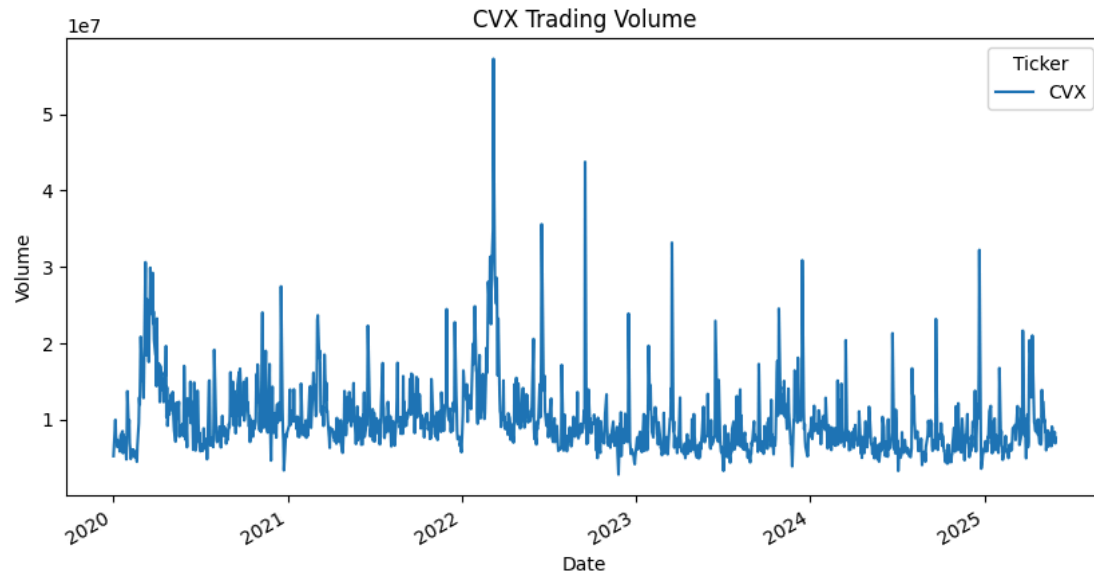


Student B: Microeconomic Specialist

Dataset Imported: Chevron, a major oil producer):

- Chevron (CVX) stock prices
- Trading volume
- Dividend payouts
- Source: Yahoo Finance using yfinance

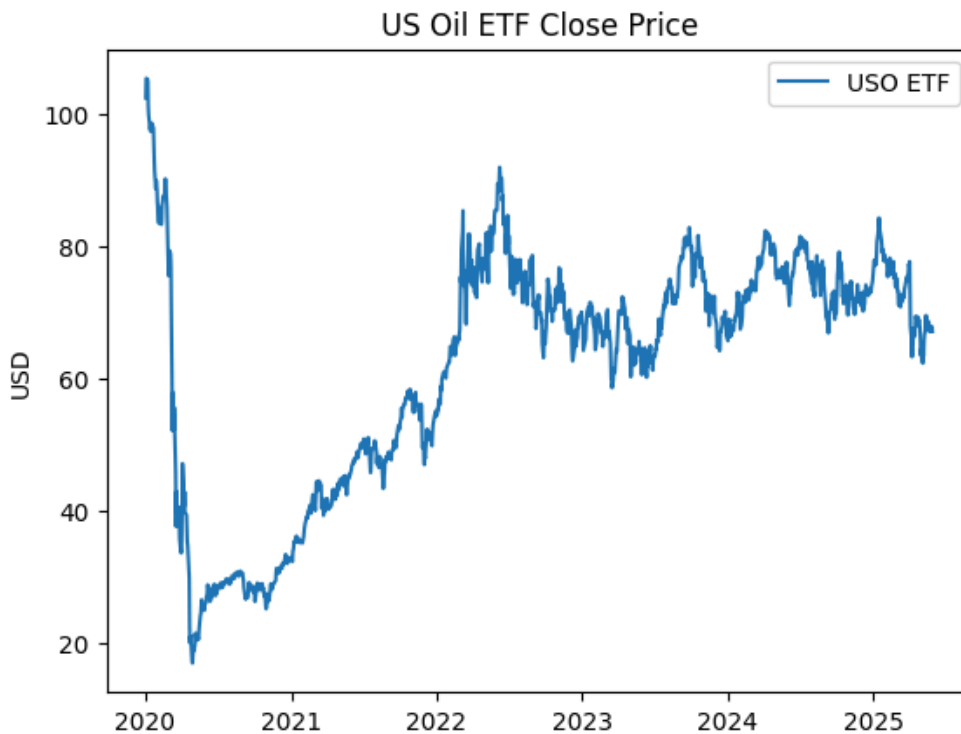




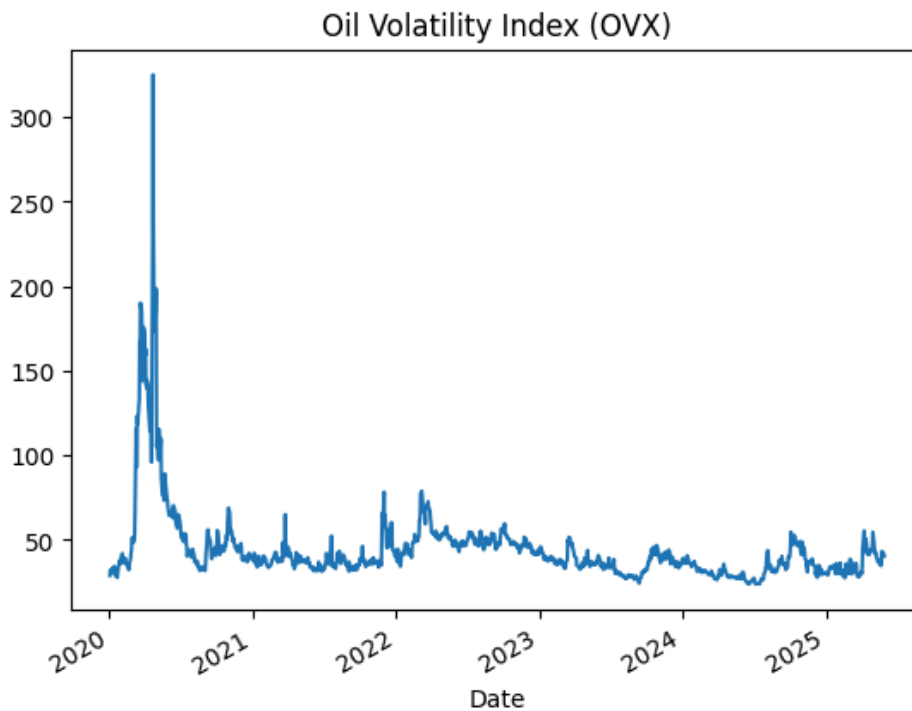
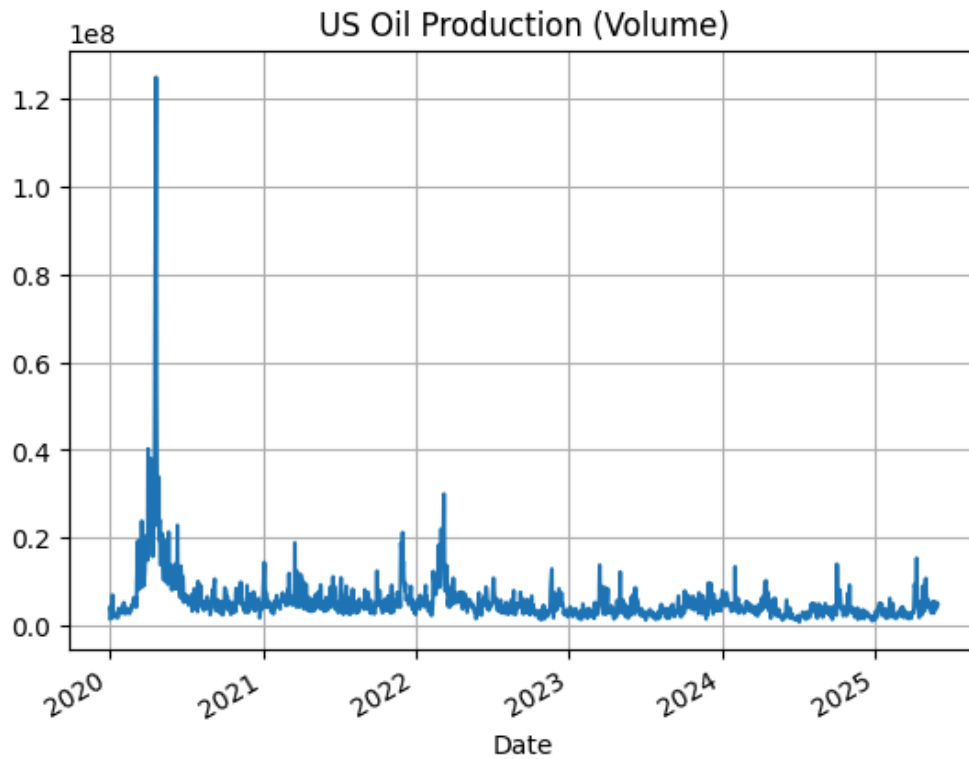
Student C: Financial Specialist

Dataset Imported:

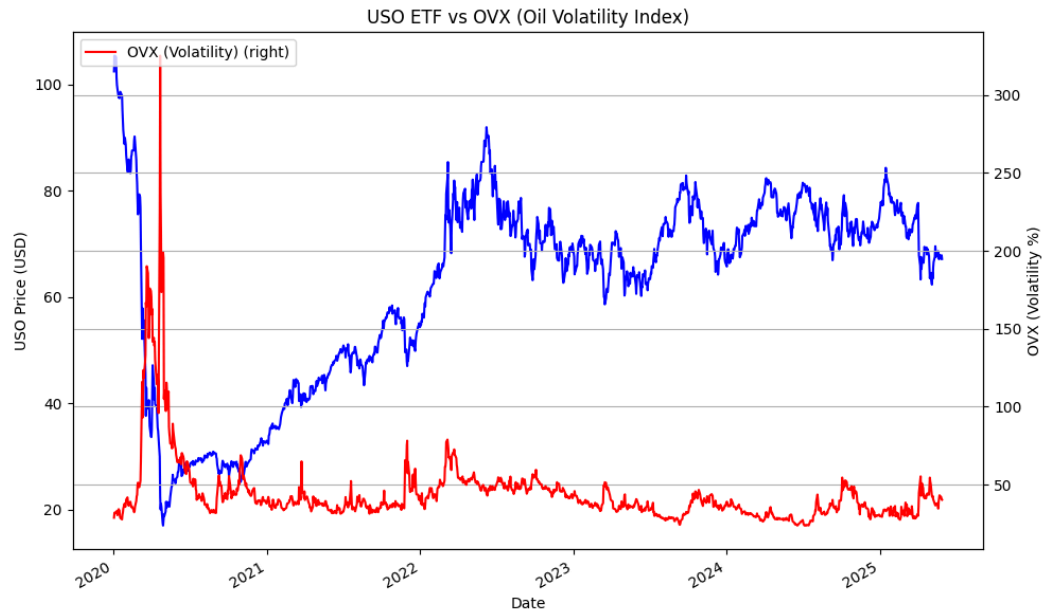
- USO ETF price data, Oil Volatility Index
- Source: Yahoo Finance



Sudden price drop between 2020-2021 and then rising trend. This is the period just after COVID-19 period. This would be due to pressure generated due to selling by investors.

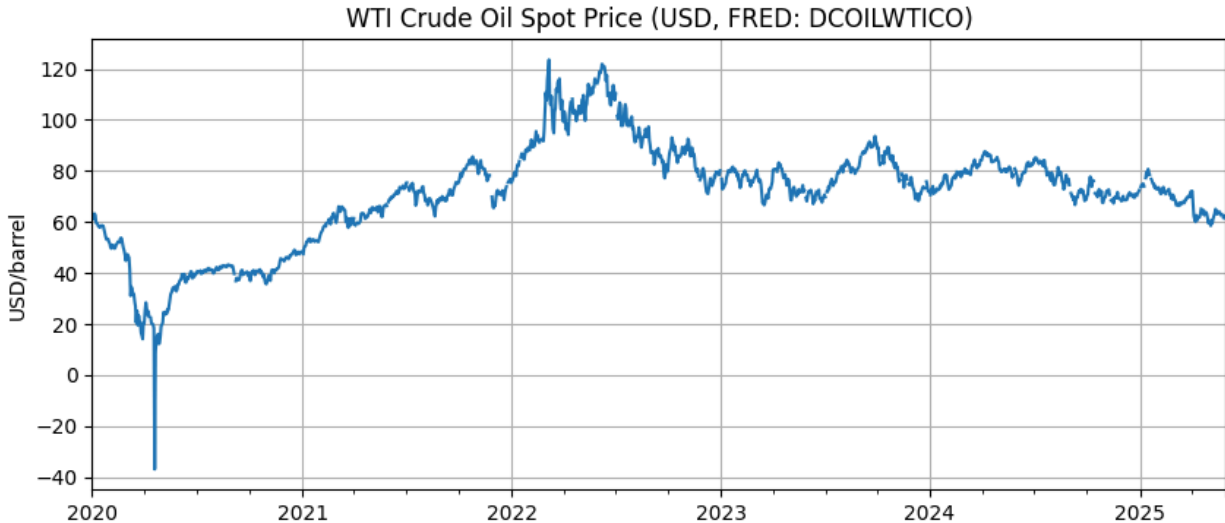


We can see high volatility and high trading volume around 2020-2021 and then returning to normalcy.



We can see that both the USO ETF and Oil WTI have some data at the beginning which may fall under outlier range.

- Source: DCOILWTICO = Daily Cushing Crude Oil, WTI spot price (EIA-derived, published by FRED)



Step 4

4.1 As a group, the team writes a dictionary of the data used and a table showing the data, frequency, source, start date, end date, and other relevant fields

Variable	Description	Frequency
GDP	Real Gross Domestic Product (US)	Quarterly
CPI	Consumer Price Index for All Urban Consumers	Monthly
stock_data	Daily stock price and volume data for Chevron (CVX)	Daily
dividends	Dividend payout history for Chevron (CVX)	Irregular (when dividends are issued)
usoil	U.S. Weekly Crude Oil Field Production	Weekly
wti	WTI Crude Oil Spot Price (Cushing, OK)	Daily (business days)
ovx	CBOE Crude Oil Volatility Index (OVX)	Daily

Start Date	End Date	Source
2020-01-01	2025-05-30	FRED (Federal Reserve Economic Data)
2020-01-01	2025-05-30	FRED
2020-01-01	2025-05-30	Yahoo Finance
2020-01-01	2025-05-30	Yahoo Finance
2020-01-01	2025-05-30	Yahoo Finance
2020-01-01	2025-05-30	FRED (Series: DCOILWTICO)
2020-01-01	2025-05-30	Yahoo Finance

Unit	Notes
Billions of Chained 2012 Dollars	Macroeconomic indicator of overall economic activity.
Index 1982-1984=100	Measures inflation experienced by consumers.
USD	Includes Open, High, Low, Close, Adjusted Close, Volume.
USD/share	Used for company-level microeconomic analysis.
Thousand Barrels per Day	Used for tracking domestic oil output.
USD/barrel	Benchmark U.S. crude oil spot price, used to approximate
Volatility index level	Implied volatility of crude oil, based on USO options; a

Step 5

5.1 Student A focuses on the “extreme outlier” part of cleaning. That is, they will identify values that are extreme compared to the rest of the data.

- Based on z-score > 3

```
#####Outliers:#####
```

```
GDP Outliers:0
```

```
CPI Outliers:0
```

```
Stock Price Outliers:0
```

```
Dividend Outliers:0
```

```
US Oil Production Outliers:16
```

```
Oil VX Outliers:131
```

```
Oil WTI Outliers:2
```

```
#####
```

- Based on IQR:

```
q1 = series.quantile(0.25)
```

```
q3 = series.quantile(0.75)
```

```
iqr = q3 - q1
```

```
outliers=[(series < q1 - 1.5 * iqr) | (series > q3 + 1.5 * iqr)]
```

```
##### Outliers based on IQR#####
```

```
Number of GDP Outliers: 0
```

```
Number of CPI Outliers: 0
```

```
Number of Stock Price Outliers: 0
```

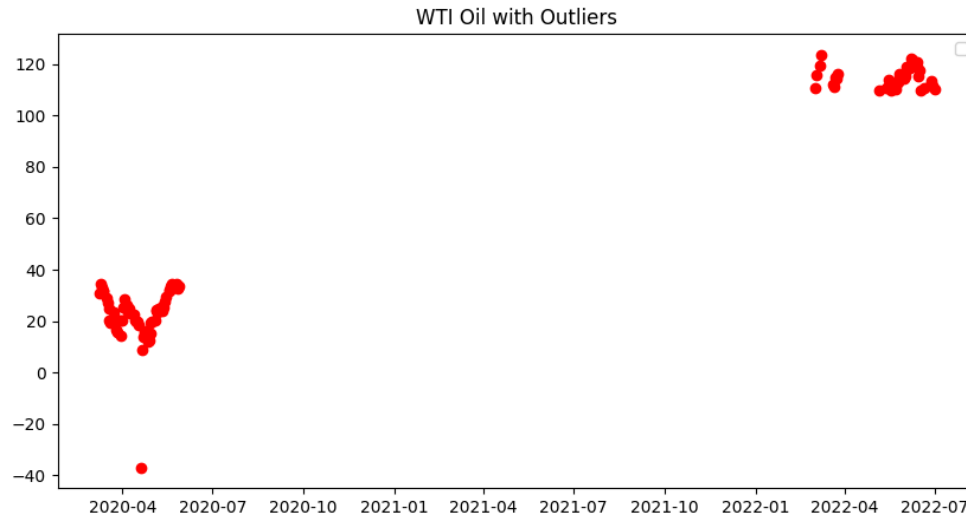
```
Number of Dividend Outliers: 0
```

```
Number of Oil Production Outliers: 1359
```

```
Number of OVX Outliers: 131
```

```
Number of WTI Oil Outliers: 97
```

```
#####
```



5.2. Student B focuses on the “bad data” part of cleaning.

No bad data like string for integers or dates in strings for certain rows. No abnormal value that is above the normal observable range limit. Duplicates not found in any variables.

5.3. Student C focuses on the “missing values” part of cleaning.

No missing values before joining the individual series.

Step 6

Once we removed the outliers from individual variable series and joined them, since the data is of varying frequency as mentioned above in Step 4. This created some NAN entries.

Joined dataset

	GDP	CPI	CVX_Stock_Close	Dividends	USOil_Production \
2020-01-01	21727.657	259.127	NaN	NaN	NaN
2020-01-02	NaN	NaN	95.001526	NaN	102.480003
2020-01-03	NaN	NaN	94.672966	NaN	105.440002
2020-01-06	NaN	NaN	94.352196	NaN	105.279999
2020-01-07	NaN	NaN	93.147369	NaN	105.040001

	Oil_VX	Oil_WTI
2020-01-01	NaN	NaN
2020-01-02	28.750000	61.17
2020-01-03	31.920000	63.00
2020-01-06	32.320000	63.27
2020-01-07	31.799999	62.70

We filled the missing values with the first row as min value and then forwardfill on the joined dataset esp. after removing outliers (using IQR) and duplicates. We have already used interpolation (linear) to fill Close price data for variable(s) series at individual level before joining them.

After Cleaning the joined dataset:

#####Cleaned Dataset : first 5 rows#####

	GDP	CPI	CVX_Stock_Close	Dividends	USOil_Production \
Date					
2020-01-01	21727.657	259.127	42.915165	1.29	17.040001
2020-01-02	21727.657	259.127	95.001556	1.29	102.480003
2020-01-03	21727.657	259.127	94.672966	1.29	105.440002
2020-01-04	21727.657	259.127	94.672966	1.29	105.440002
2020-01-05	21727.657	259.127	94.672966	1.29	105.440002

	Oil_VX	Oil_WTI
Date		
2020-01-01	23.73	35.49
2020-01-02	28.75	61.17
2020-01-03	31.92	63.00
2020-01-04	31.92	63.00
2020-01-05	31.92	63.00

#####

```
#### Null Values#####
```

```
GDP          0
CPI           0
CVX_Stock_Close  0
Dividends     0
USOil_Production  0
Oil_VX        0
Oil_WTI       0
```

```
dtype: int64
```

```
#####
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
DatetimeIndex: 1977 entries, 2020-01-01 to 2025-05-30
```

```
Freq: D
```

```
Data columns (total 7 columns):
```

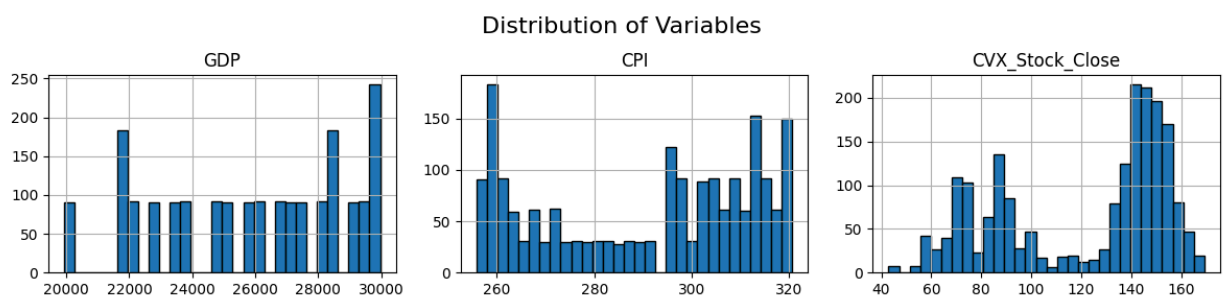
#	Column	Non-Null Count	Dtype
0	GDP	1977 non-null	float64
1	CPI	1977 non-null	float64
2	CVX_Stock_Close	1977 non-null	float64
3	Dividends	1977 non-null	float64
4	USOil_Production	1977 non-null	float64
5	Oil_VX	1977 non-null	float64
6	Oil_WTI	1977 non-null	float64

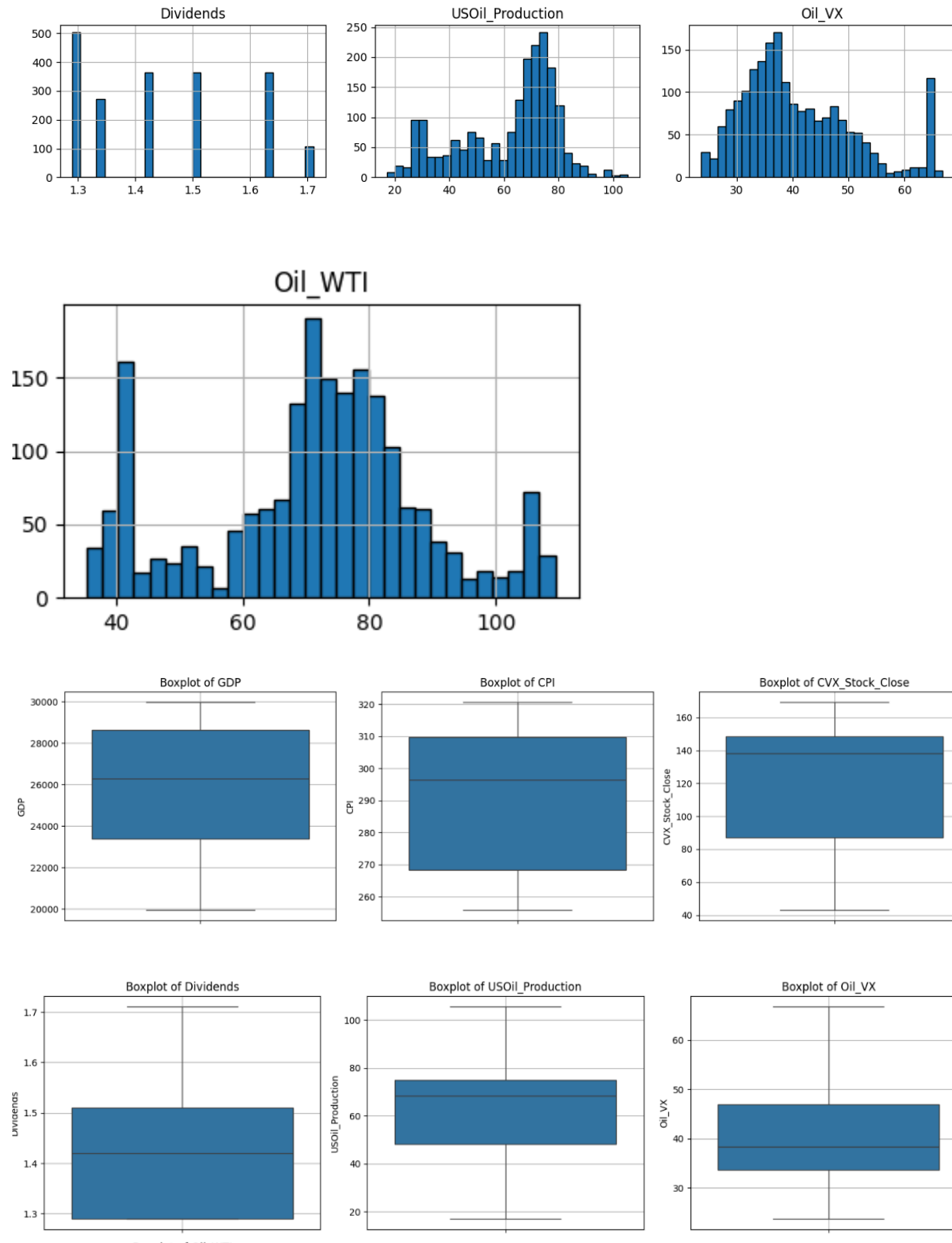
```
dtypes: float64(7)
```

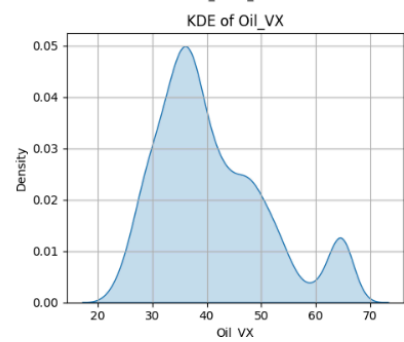
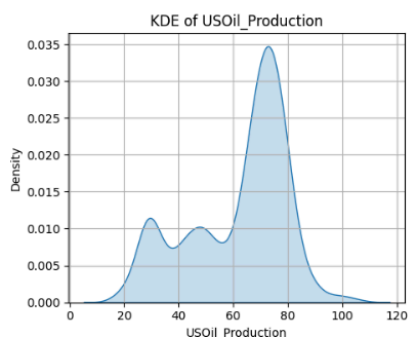
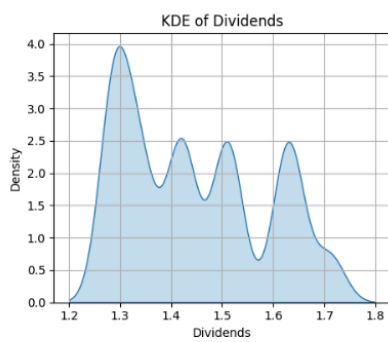
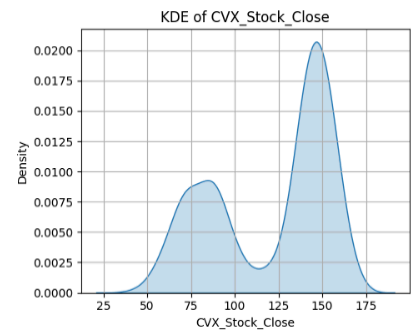
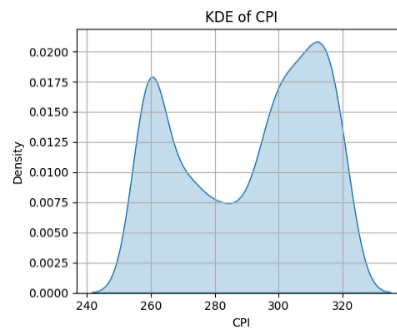
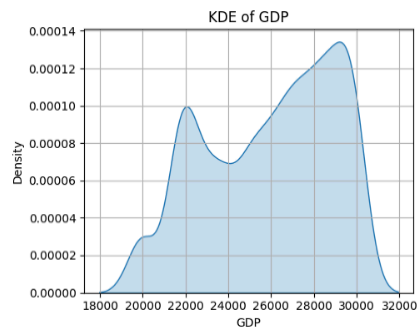
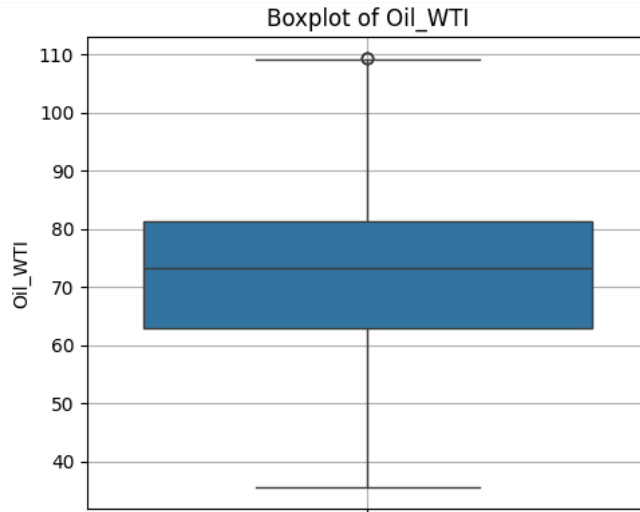
Step 7

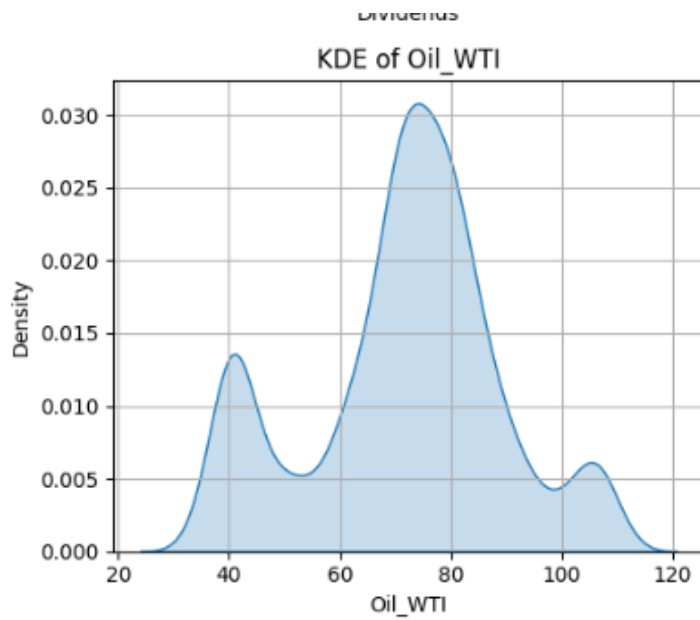
7.a. Student A creates “distributional” plots.

Histograms, BoxPlots, KDE Plots



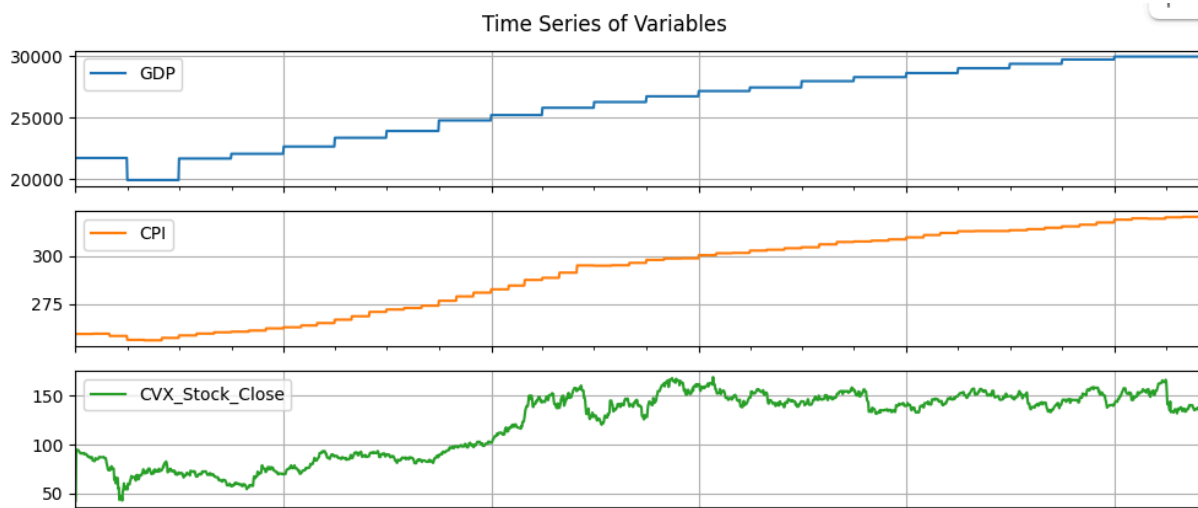


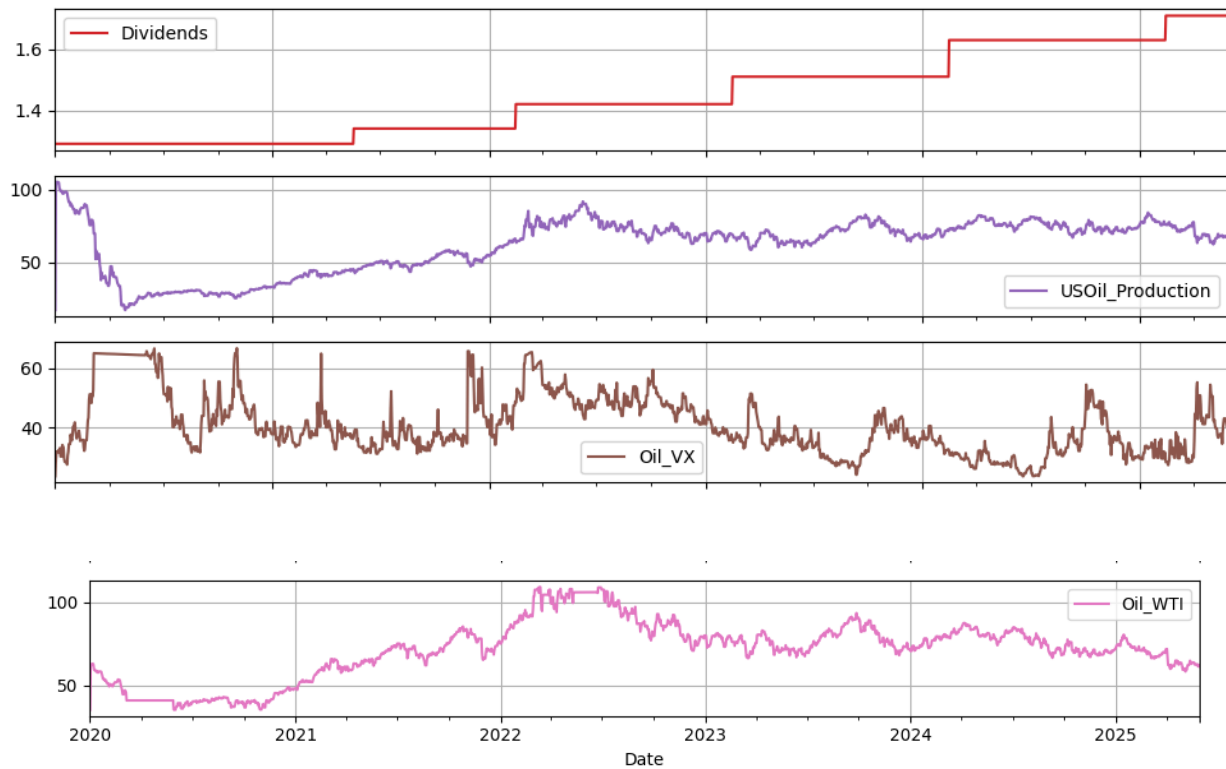




Almost multimodal distribution may mean clustering or regimes e.g. low volatility, medium volatility regime or high volatility regimes.

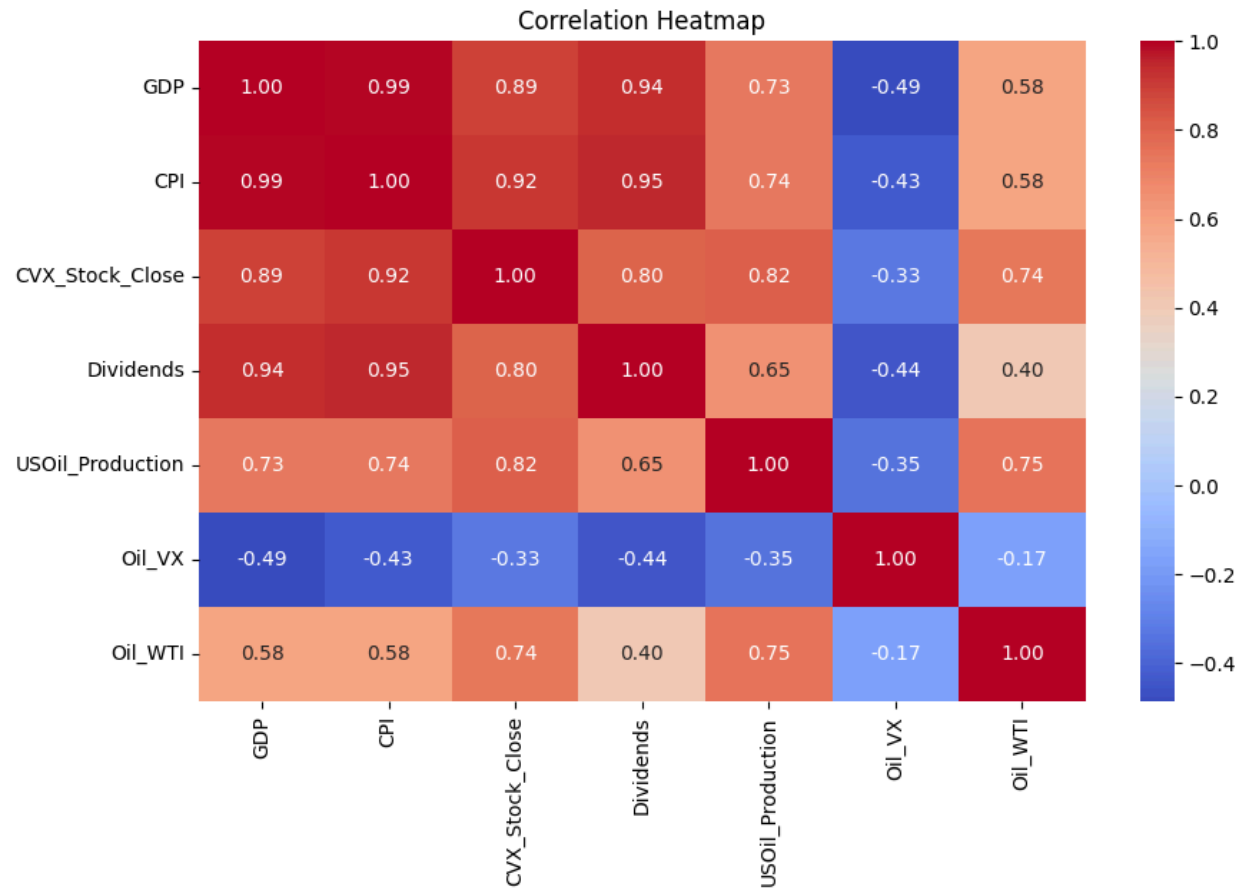
7.b. Student B creates "time series" plots.



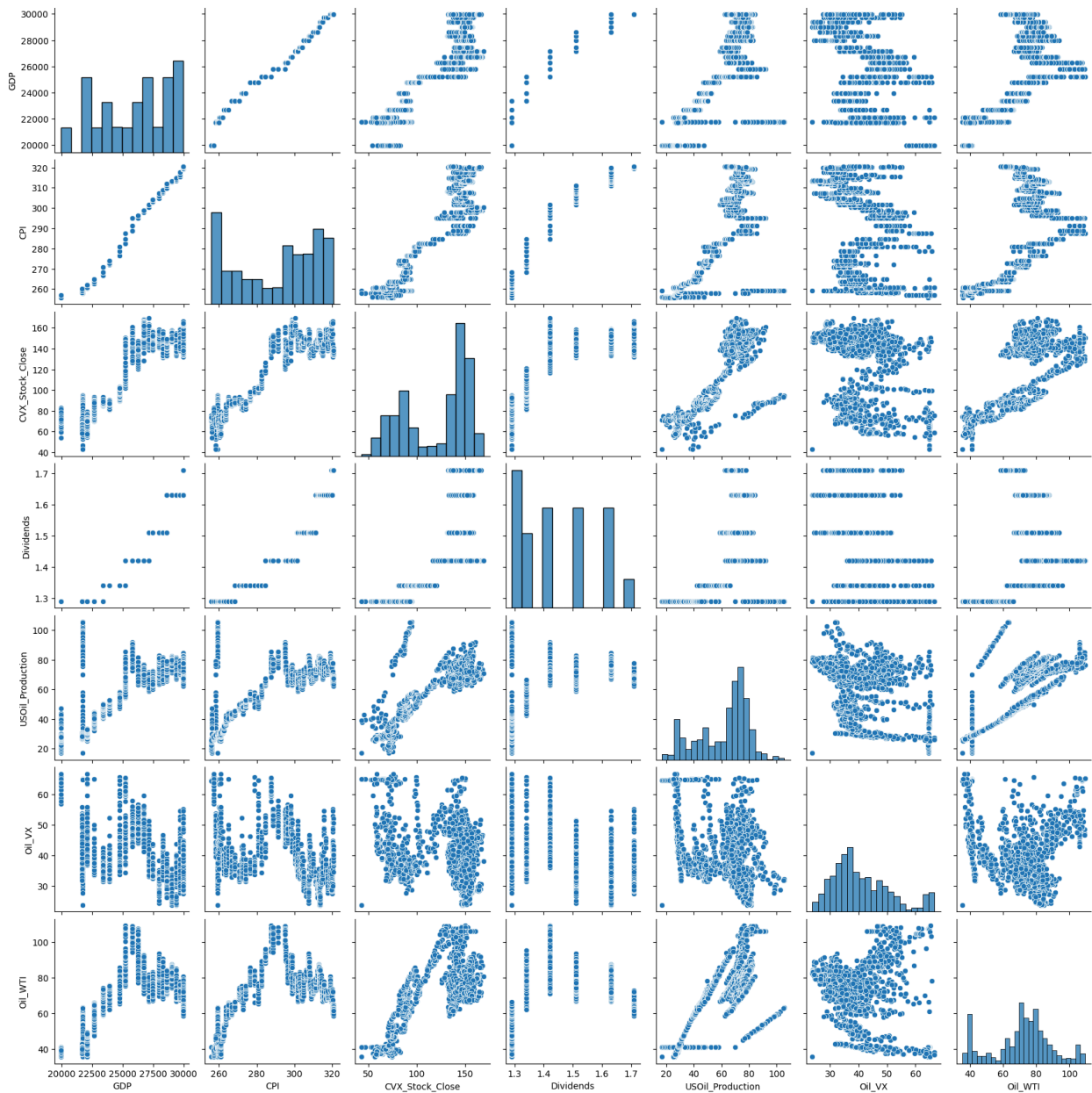


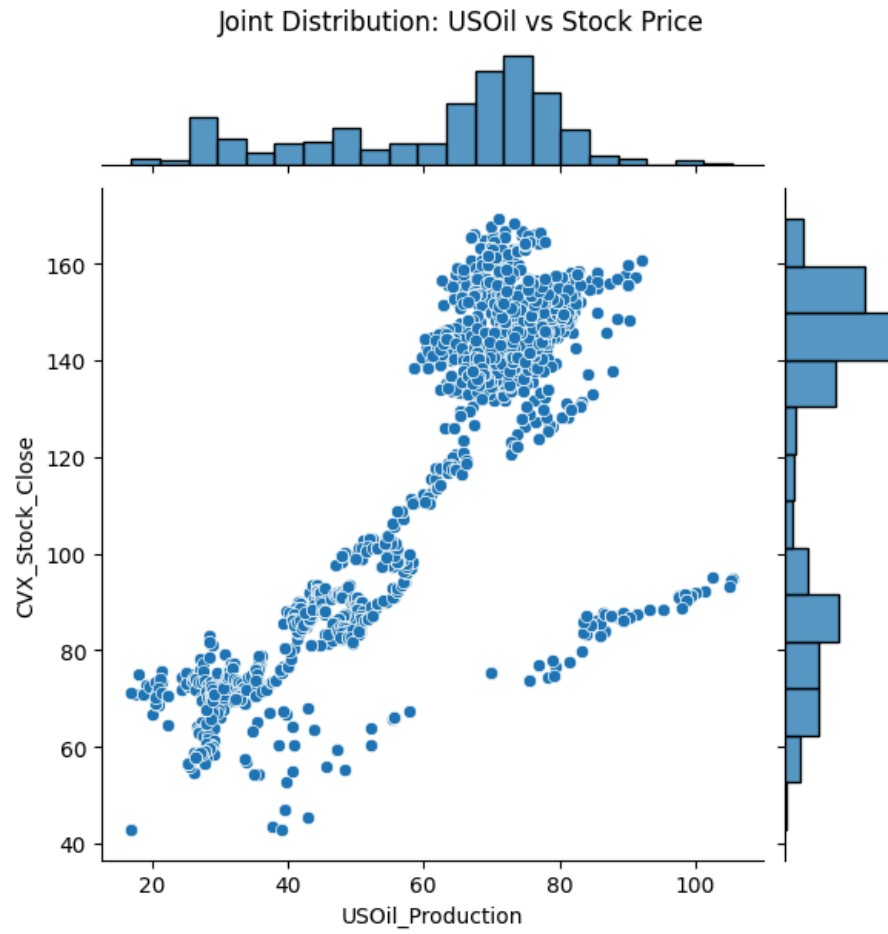
7.c. Student C creates “multivariate” plots (distributional and/or time series)

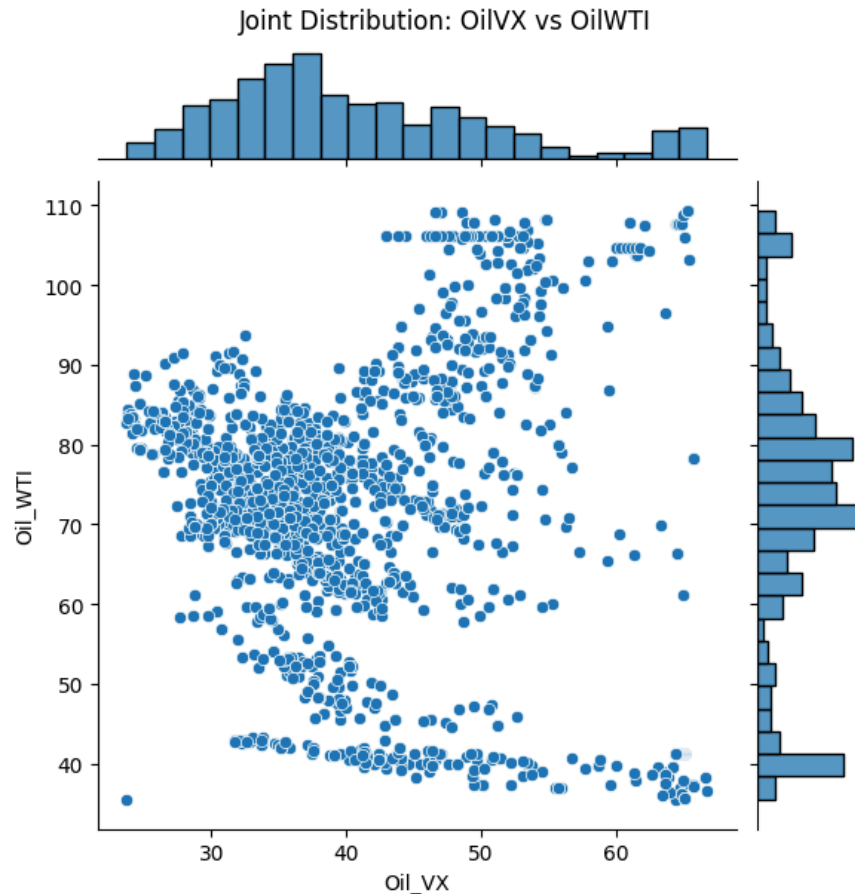
Correlation Heatmap, Pairplot, Jointplot



Pairwise Relationships







Step 8

8.a. What makes oil prices look different from other asset prices? E.g., spikes, clustered volatility, seasonality, etc.

- **Spikes & Shocks:** The oil price data (e.g., from USO ETF) shows sharp, sudden spikes and drops that are more frequent and extreme than in typical equity prices. These reflect geopolitical shocks, OPEC decisions, or global crises (like COVID-19 in early 2020).
- **Clustered Volatility:** Volatility appears in bursts, with low volatility periods followed by high-volatility episodes and then back to low volatility. This is evident in the outlier detection code and visual line plots.
- **Seasonality:** Though not deeply explored in the notebook, energy prices often show seasonal trends, especially around winter or travel-heavy months due to demand cycles.

- **Mean-Reversion Tendency:** Unlike some stocks that trend long-term, oil prices often revert back to a certain range due to supply-side adjustments (e.g., production cuts) or when the situations like war in the Middle East returns to normalcy.
- Multi modal distribution could mean clustering or regime. The modals of the plots are not same or similar, different spikes or trends have happened at different time intervals.

8.b. What types of distributions do oil returns have?

- Oil returns (e.g., from the USO ETF) exhibit non-normal, heavy-tailed distributions:
 - Presence of fat tails (extreme returns occur more often than in a normal distribution).
 - Likely skewed, depending on market events.
- The notebook shows kernel density plots and scatterplots showing outliers and asymmetry in the return distribution. For e.g. Oil WTI spot price and Oil Volatility index (OVX) have outliers like USO.
- These characteristics are consistent with stylized facts of financial returns and suggest that oil returns may follow a Lévy-stable or t-distribution rather than Gaussian.

8.c. What type of autocorrelation do the oil returns have?

- Though autocorrelation plots (acf, pacf) aren't shown directly in the notebook, typical behavior includes:
 - **Low autocorrelation in returns:** Monthly if not Daily oil returns themselves are usually uncorrelated, suggesting weak predictability in direction.
 - **High autocorrelation in volatility:** The magnitude of returns (absolute or squared) exhibits strong autocorrelation, consistent with volatility clustering.

- This would suggest that models like GARCH or stochastic volatility would be appropriate for modeling oil returns.

8.d. What other stylized facts can you say about oil prices.

- **Asymmetry:** Oil prices often fall faster than they rise, especially during supply-demand shocks (e.g., war, pandemics, recessions).
- **Non-linearity:** Relationships with macro variables (like GDP or CPI) are often nonlinear and regime-dependent.
- **Jumps and Gaps:** Due to overnight geopolitical developments, oil prices show jumps rather than smooth transitions.
- **Market Reaction to Futures:** As noted in the thesis, futures prices often lead spot prices, showing that expectations are priced in early.

Step 9

9.a. Student A defines probabilistic graphic models. Be sure to distinguish belief networks and Markov networks.

Probabilistic Graphical Models, or PGMs for short, are basically a type of model that combines graph theory with probability to help us understand complex systems—especially when there's uncertainty involved, which is pretty much always the case in real-world problems. You see them used in all sorts of fields like forecasting energy usage, healthcare, finance... anywhere where decisions need to be made without having all the data or when the data is noisy or not super reliable.

In the case of oil price forecasting, PGMs are especially helpful because they let us model how different economic, political, and market-related variables interact with each other. For example, we might want to see how U.S. GDP, inflation, or decisions made by OPEC can affect oil prices. PGMs give us a structure that we can actually interpret and work with—not just numbers on a spreadsheet, but something that shows how variables are connected. And on top of that, they can also be used for making predictions or running different “what if” scenarios.

At the heart of it, a PGM is just a graph. Each node in the graph is a random variable, like interest rates or oil inventories, and the edges between them represent probabilistic dependencies. What makes this really useful is that instead of trying to work with one giant probability

distribution that covers every variable at once, which gets out of hand fast, PGMs let us break that down into smaller, more manageable parts—based on how variables are conditionally independent from each other.

There are mainly two types of PGMs. One type uses directed graphs, and the other one uses undirected graphs. These are known as Bayesian Belief Networks and Markov Networks, respectively. Each has its own strengths depending on what you're trying to model.

Bayesian Belief Networks (BBNs)

- These are directed acyclic graphs (DAGs).
- Each node is a random variable, and edges imply causal or conditional dependence.
- The joint distribution is factorized as a product of conditional probabilities:
$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Parents}(X_i))$$
- BBNs are highly interpretable and allow for causal reasoning.
- Example in oil markets:
Interest Rate $\rightarrow$ GDP $\rightarrow$ Oil Demand $\rightarrow$ Oil Supply $\rightarrow$ Oil Price
Here, GDP influences oil demand, which in turn affects oil prices.
Inference in BBNs is done by conditioning on known variables and computing the updated probabilities across the network
- Advantages
 - Naturally captures causal relationships.
 - Supports reasoning with missing or partial data.
 - Can incorporate expert knowledge into the structure.

Markov Networks (or Markov Random Fields)

- These are undirected graphs.
- Edges represent mutual dependence (not causality).
- The distribution is using cliques and potential functions:
$$P(X_1, \dots, X_n) = 1/Z * \prod_{\text{cliques}} \phi(C_i)$$

where ϕ (phi) are potential functions and Z is a normalization constant.
- Useful for domains where symmetry and correlations dominate over causality.
- Less intuitive for causal modeling, but effective for spatial/temporal smoothing and computer vision.
- Example: In oil modeling, a Markov network might model how Oil Inventories, Global Supply, and Oil Prices co-vary without a specific causal direction.
- Advantages
 - Good for symmetrical relationships.

- Preferred when causal ordering is unknown.

Key Differences

Feature	Bayesian Networks	Markov Networks
Structure	Directed Acyclic Graph (DAG)	Undirected Graph
Focus	Causality, conditional dependence	Symmetric correlation
Factorization	Product of conditional probabilities	Product of clique potentials
Interpretation	Intuitive, causal	Correlational
Inference complexity	Easier in sparse DAGs	Often requires approximation

Probabilistic Graphical Models are a really useful way to deal with systems that are super complex and unpredictable — like the oil market. They give us a proper framework to understand how different things are connected, even when we don't have perfect data or everything is kind of messy. In the case of forecasting oil prices, Bayesian Networks make a lot of sense because they let us represent known causal links — like, say, when supply drops, prices usually go up.

What's also nice about them is that they allow you to bring in both expert intuition and actual data. So you're not relying only on numbers or only on assumptions, but finding a balance between the two. And even when some data is missing — which happens more often than not — the model can still give you something useful.

Now, Markov Networks aren't really the main focus here, but they're still worth mentioning. They're more about capturing relationships without worrying about what causes what, which can be handy when causality isn't clear or when everything seems to be influencing everything else.

Overall, PGMs give you a flexible and understandable way to approach problems like this. When you're dealing with something as uncertain and fast-moving as the global energy market, having a model that can adapt, make sense of the chaos, and still explain itself is a huge advantage.

9.b. Student B illustrates parameter learning and distinguishes it from structure learning.

In Probabilistic Graphical Models (PGMs) like Bayesian Networks, learning from data involves two major components:

1. **Structure Learning:** Discovering how variables are connected, which variable influences which.
2. **Parameter Learning:** Estimating the actual probabilities (or parameters) that quantify those relationships.

In the context of oil price forecasting, we know that OPEC decisions affect supply and that supply affects price (structure), but we still need data to estimate how strongly these effects operate (parameters).

Structure Learning:

Structure learning is about the graph, the network (or polygon) of edges and nodes, that represents the dependencies among variables.

In Bayesian Networks, it involves identifying:

- Which nodes are connected

- The direction of those connections (e.g., does GDP → Oil Price or vice versa?)

Structure learning can be done via:

- Expert knowledge (manual specification)
- Data-driven methods: Score-based (e.g., BIC, AIC), Constraint-based (e.g., PC algorithm), Hybrid methods

Structure learning helps answer: What influences what?

But it doesn't tell us the numerical strength of those relationships that's where parameter learning comes in.

Parameter Learning:

Parameter learning is about estimating the probabilities or numerical relationships between connected variables, given a known structure or estimating the Conditional Probability Tables for a Bayesian network..

For example:

$P(\text{Oil_Price} \mid \text{Supply}, \text{Demand})$

There are two major cases:

1. Known structure + complete data:

- Use Maximum Likelihood Estimation (MLE) or Bayesian Estimation
- Simple counting: E.g., if 80% of the time high supply leads to low prices, then $P(\text{Price}=\text{low} \mid \text{Supply}=\text{high})=0.8$

2. Incomplete data:

Use Expectation-Maximization (EM) to iteratively estimate missing values and parameters

Parameter learning answers:

- How likely is Y given X?
- How strong is the influence of X on Y?

Key Differences

Feature	Structure Learning	Parameter Learning
What it learns	Graph topology (edges, direction)	Probabilities for existing edges
Based on	Statistical independence, scoring metrics	Likelihood from data
Involves	Searching over graph space	Estimating CPTs / weights
Output	DAG or undirected graph	Conditional Probability Tables or factors
Difficulty	Harder, often combinatorially complex	Easier if structure is known

Application to Oil Forecasting

In the oil price forecasting project:

- Structure Learning helps discover if Interest Rate, GDP, and OPEC output all influence Oil Prices, and how they're connected.
- Parameter Learning then tells us how strong those links are — e.g., does a 1% increase in GDP increase oil prices by 0.5% or 2%?

The project uses both:

- Expert knowledge to define partial structure
- Data-driven parameter learning to quantify relationships

E.g. $P(\text{Oil_Price} = \text{high} \mid \text{Demand} = \text{high}, \text{Supply} = \text{low}) = 0.82$

i.e. when we have situation like demand is high and supply is low, it is predicted that there's an 82% chance that oil price will be high.

Component	Purpose
Structure	Learn who influences whom
Parameters	Learn how strong that influence is

If **Structure** is the skeleton of the model, the **Parameters** give us the muscle that makes predictions

In oil price modeling, getting the structure right ensures we're asking the **right questions**, and parameter learning ensures we're answering them with **data-backed precision**.

9.c. Student C describes Markov chains and Markov blankets

In the study of probabilistic graphical models and systems that evolve over time, like financial markets or oil prices, two concepts often come up that help us deal with all that complexity: Markov Chains and Markov Blankets. Both are grounded in the idea of conditional independence, which basically means that if we know certain variables, we can kind of ignore others when making predictions. This idea becomes really useful when we're working in high-dimensional or noisy environments like the energy markets, where not everything is always available or clean.

Markov Chains: Modeling Sequential Dependency

A **Markov Chain** is a model of a system that transitions from one state to another in **discrete time steps**, where the probability of moving to the next state **depends only on the current state**.

$$P(X_t | X_{t-1}, X_{t-2}, \dots, X_0) = P(X_t | X_{t-1})$$

Key Components:

- States: Possible values (e.g., oil price levels: low, medium, high)
- Transition matrix: A matrix of probabilities between states
- Initial distribution: The starting probability over states

Relevance to Oil Forecasting:

Markov Chains are useful for modeling how **oil price regimes evolve over time** for example, how a high-price or high volatility regime might shift into a low price or lower volatility regime depending on demand or geopolitical shocks.

Example:

- Model daily transitions in volatility regimes
- Estimate probability of moving from low volatility to high volatility tomorrow

Feature	Description
Time structure	Discrete steps
Memory	Only depends on current state (Markov property)
Visualization	State diagram (circles + arrows)
Common applications	Regime switching, volatility modeling, games

Markov Blanket: The Minimal Conditioning Set

A Markov Blanket of a node in a probabilistic graphical model (like a Bayesian Network) is the minimal set of nodes that shield it from the rest of the network.

In a Bayesian Network, the Markov Blanket of a node **X** includes:

- Its parents (direct causes)
- Its children
- Its children's other parents (co-parents)

If we know the values of the Markov Blanket of X, then X is conditionally independent of all other variables in the network.

$$P(X | \text{All other variables}) = P(X | \text{Markov Blanket of X})$$

Relevance to Oil Forecasting:

- If oil price is the node of interest, its Markov blanket might include:
 - Oil demand and supply (parents)
 - Energy index and stock prices (children)
 - Macroeconomic factors like GDP (co-parents)
- Once we know this set, we can ignore everything else when modeling or predicting oil prices, making inference much faster and models more interpretable.

In a Bayesian network where:

Interest_Rate → GDP → Oil_Demand → Oil_Price ← Supply

The Markov Blanket of Oil_Price is:

- Oil_Demand (parent)
- Supply (parent)
- Any child of Oil_Price (e.g., inflation) and their other parents

Comparison

Concept	Focus Area	Purpose
Markov Chain	Time series modeling	Describe how states evolve sequentially
Markov Blanket	Graph-based inference	Isolate relevant variables for prediction

Markov Chains and Markov Blankets are two core ideas that play a big role in simplifying and understanding complex probabilistic models. Markov Chains help us capture how systems change over time, which makes them especially useful when working with time series data, like oil prices or shifts in market volatility. On the other hand, Markov Blankets help us figure out the minimum set of variables we need to focus on when making predictions. Instead of having to consider the entire network, we can just look at the most directly related factors and ignore the rest.

In the context of oil price forecasting, both of these concepts are incredibly useful. They allow us to build models that are not only scalable and easier to interpret, but also efficient from a computational point of view. And perhaps most importantly, they still allow the model to reason under uncertainty, which is exactly what we need when dealing with real-world markets.

Step 10

10.1 As a group, the team writes their own pseudocode version of Algorithm1: Inferred Causality.

Algorithm 1:

To identify causal relationships among variables by:

- Using structure learning to discover dependencies,
- Using Markov blankets to limit conditioning sets,
- And estimating parameters to quantify causal strength.

Algorithm: Inferred_Causality_Graph_Model(Data)

Input:

Data - a dataset with n variables and m observations

Output:

G - a directed graph (causal structure)

Θ - parameters (conditional probabilities)

Steps:

1. Initialize:

Let G = empty directed acyclic graph

Let VariableSet = $\{X_1, X_2, \dots, X_n\}$

2. Structure Learning (Identify Candidate Causal Links):

For each variable X_i in VariableSet:

For each variable $X_j \neq X_i$:

Test conditional independence between X_i and X_j

using a statistical test (e.g., chi-square, mutual
info)

over possible subsets of conditioning variables

If X_i and X_j are not conditionally independent:

Add edge $(X_j \rightarrow X_i)$ to graph G (provisional)

3. Prune using Markov Blankets:

For each variable X_i :

Identify $\text{MarkovBlanket}(X_i) = \text{Parents}(X_i) \cup \text{Children}(X_i)$

$\cup$ Co-parents

For any variable X_j not in $\text{MarkovBlanket}(X_i)$:

Remove edge $X_j \rightarrow X_i$ if present

4. Parameter Learning:

For each node X_i in G :

Estimate conditional probability table (CPT)

$P(X_i \mid \text{Parents}(X_i))$ using Maximum Likelihood Estimation
or Expectation-Maximization if data has missing values

5. Output:

Return graph G with directed edges representing inferred
causality

and $\Theta = \{P(X_i \mid \text{Parents}(X_i)) \text{ for all } X_i\}$ as learned parameters

Application: Oil Price Forecasting

- Variables: Oil_Price, GDP, CPI, Inventory, Supply_Shocks
- The algorithm learns:
 - Structure: e.g., $\text{GDP} \rightarrow \text{Oil_Price} \leftarrow \text{Inventory}$
 - Markov Blanket: for Oil_Price, includes GDP and Inventory
 - Parameters: e.g., $P(\text{Oil_Price}=\text{high} \mid \text{GDP}=\text{low}, \text{Inventory}=\text{low})=0.87$

References:

1. Koller, Daphne, and Nir Friedman. *Probabilistic Graphical Models: Principles and Techniques*. MIT Press, 2009.
July 31, 2009. <https://mitpress.mit.edu/9780262013192/probabilistic-graphical-models/>.
2. Pearl, Judea. *Causality: Models, Reasoning, and Inference*. 2nd ed., Cambridge University Press, 2009.
3. Murphy, Kevin P. *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012.
4. Jensen, Finn V., and Thomas D. Nielsen. *Bayesian Networks and Decision Graphs*. 2nd ed., Springer, 2007.
5. Lauritzen, Steffen L. *Graphical Models*. Oxford University Press, 1996.
6. Alquist, Ron, and Lutz Kilian. "What Do We Learn from the Price of Crude Oil Futures?" *Journal of Applied Econometrics*, vol. 25, no. 4, June 2010, pp. 539–573.
<https://onlinelibrary.wiley.com/doi/abs/10.1002/jae.1159>.
7. Kilian, Lutz, and Ron Alquist. "What Do We Learn from the Price of Crude Oil Futures?" CEPR Discussion Paper No. 6548, Nov. 2007.

GROUP WORK PROJECT # _1_
Group Number: __10177__

MScFE 660: RISK MANAGEMENT

<https://cepr.org/publications/dp6548>.

8. Hamilton, James D. Time Series Analysis. Princeton University Press, 1994.